MTH 201 Chapter 4: Power Series

4.3 Representing Functions as Power Series

Motivation: Consider the series $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$. We saw earlier that this a geometric series with interval of convergence: $(-1,1)$.

From section 3.2 we know that if $|x| < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. This means that on the interval $(-1,1)$ we can represent the function $f(x) = \frac{1}{1-x}$ as a power series.

Note: A good question to ask at this point would be: Why do we care if we can represent certain functions as a power series? As it turns out, being able to represent functions as power series is tremendously important to many areas of both pure and applied mathematics. In particular, this can be useful for solving differential equations, approximating functions by polynomials, and integration of functions which don't have an elementary antiderivative.

Example 1: Find a power series representation for $g(x) = \frac{1}{2+x}$ and state its interval of convergence.

We want to relate this function to the geometric series sum formula. We calculate that

$\frac{1}{2+x} = \frac{1}{2\left(1+\frac{x}{2}\right)} = \frac{1/2}{\left(1-\left(-\frac{x}{2}\right)\right)}$. We know that this is the sum of the geometric series with $a = 1/2$ and $r = \frac{-x}{2}$ provided that $\left|\frac{-x}{2}\right| < 1 \Leftrightarrow |-x| < 2 \Leftrightarrow -2 < x < 2$. This means that a power series representation for the function is: $\sum_{n=0}^{\infty} \left(\frac{1}{2}\left(\frac{-x}{2}\right)^n\right)$ and its interval of convergence is $(-2,2)$.

Example 2 (you try) : Find a power series representation of the function $f(x) = \frac{x}{2x^2+1}$ and state its interval of convergence.

We calculate that $\frac{x}{2x^2+1} = \frac{x}{1-\left(-2x^2\right)}$. We know that this is the sum of the geometric series with $a = x$ and $r = -2x^2$ provided that $\left|-2x^2\right| < 1 \Leftrightarrow x^2 < \frac{1}{2} \Leftrightarrow \frac{-\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$.

This means that a power series representation for the function is: $\sum_{n=0}^{\infty} \left(x\left(-2x^2\right)^n\right) = \sum_{n=0}^{\infty} \left((-2)^n x^{2n+1}\right)$ and its interval of convergence is $\left(\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$.

Example 3: Find a power series representation for $g(x) = \frac{1}{(x-1)(x-3)}$. Find its interval of convergence.

We will first use partial fractions to write $g(x)$ as a sum of two fractions.

We know that $g(x) = \frac{1}{(x-1)(x-3)} = \frac{A}{(x-1)} + \frac{B}{(x-3)}$. It must be the case that $1 = A(x-3) + B(x-1)$.

If $x = 1$, then $1 = -2A \Rightarrow A = -\frac{1}{2}$

If $x = 3$, then $1 = 2B \Rightarrow B = \frac{1}{2}$

Therefore $g(x) = \frac{1}{(x-1)(x-3)} = \frac{-1/2}{(x-1)} + \frac{1/2}{(x-3)} = \frac{1/2}{1-x} + \frac{-1/2}{3-x} = \frac{1/2}{1-x} - \frac{1/6}{1-\frac{x}{3}}$

Each fraction represents the sum of a geometric series. Specifically, $\frac{1/2}{1-x} = \sum_{n=0}^{\infty} \frac{1}{2} x^n$ for $-1 < x < 1$ and

$\frac{1/6}{1-\frac{x}{3}} = \sum_{n=0}^{\infty} \frac{1}{6}\left(\frac{x}{3}\right)^n$ for $\left|\frac{x}{3}\right| < 1 \Rightarrow -1 < \frac{x}{3} < 1 \Rightarrow -3 < x < 3$.

Therefore if we use values of *x* in the interval $(-1,1)$ both series will converge and we can say that

$$g(x) = \frac{1}{(x-1)(x-3)} = \frac{1/2}{1-x} - \frac{1/6}{1-\frac{x}{3}} = \sum_{n=0}^{\infty} \frac{1}{2} x^n - \sum_{n=0}^{\infty} \frac{1}{6}\left(\frac{x}{3}\right)^n = \sum_{n=0}^{\infty}\left[\frac{1}{2} x^n - \frac{1}{6}\left(\frac{x}{3}\right)^n\right]$$

$$= \sum_{n=0}^{\infty}\left[\frac{1}{2} x^n - \frac{1}{6} \cdot \frac{1}{3^n} x^n\right] = \sum_{n=0}^{\infty}\left(\frac{1}{2} - \frac{1}{6 \cdot 3^n}\right) x^n$$

Note: Other operations such as multiplication and division can be done with power series, although they are a bit more complicated. We will not discuss these ideas in class, but details of these operations are discussed in the book.

Theorem (Addition & Subtraction of Power Series): Suppose that the power series $\sum_{n=0}^{\infty} c_n (x-x_0)^n$ and $\sum_{n=0}^{\infty} d_n (x-x_0)^n$ converge to the functions *f* and *g*, respectively on a common interval *I* (centered at x_0). Then, the power series $\sum_{n=0}^{\infty} (c_n + d_n)(x-x_0)^n$ converges to the function $f+g$ on *I*.

Proof: (NOT DONE IN CLASS)

Suppose that the power series $\sum_{n=0}^{\infty} c_n\left(x-x_0\right)^n$ and $\sum_{n=0}^{\infty} d_n\left(x-x_0\right)^n$ converge to the functions *f* and *g*, respectively on a common interval *I* (centered at x_0).

Let *x* be some point in *I* and let $S_N\left(x\right)$ and $T_N\left(x\right)$ denote the *N*th partial sums of the series respectively. Then, the sequence $\left\{S_N\left(x\right)\right\}$ converges to $f\left(x\right)$ and the sequence $\left\{T_N\left(x\right)\right\}$ converges to $g\left(x\right)$.

We also know that the *N*th partial sum of $\sum_{n=0}^{\infty}\left(c_n+d_n\right)\left(x-x_0\right)^n$ is:

$$\sum_{n=0}^{N}\left(c_n+d_n\right)\left(x-x_0\right)^n=\sum_{n=0}^{N}\left(c_n\left(x-x_0\right)^n+d_n\left(x-x_0\right)^n\right)=\sum_{n=0}^{N}c_n\left(x-x_0\right)^n+\sum_{n=0}^{N}d_n\left(x-x_0\right)^n=S_N\left(x\right)+T_N\left(x\right)$$

Therefore we see that $\lim_{N\to\infty}\left[S_N\left(x\right)+T_N\left(x\right)\right]=\lim_{N\to\infty}S_N\left(x\right)+\lim_{N\to\infty}T_N\left(x\right)=f\left(x\right)+g\left(x\right)$ and so we conclude that $\sum_{n=0}^{\infty}\left(c_n+d_n\right)\left(x-x_0\right)^n$ converges to $f+g$ on *I*.

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Theorem: If the power series $\sum_{n=0}^{\infty} c_n\left(x-x_0\right)^n$ has radius of convergence $R>0$, then the function f defined by $f(x)=\sum_{n=0}^{\infty} c_n\left(x-x_0\right)^n$ is differentiable on the interval $\left(x_0-R, x_0+R\right)$ and

1. $f'(x)=c_1+2c_2\left(x-x_0\right)+3c_3\left(x-x_0\right)^2+\cdots=\sum_{n=1}^{\infty} nc_n\left(x-x_0\right)^{n-1}$
2. $\int f(x)dx=C+c_0\left(x-a\right)+c_1\dfrac{\left(x-x_0\right)^2}{2}+c_2\dfrac{\left(x-x_0\right)^3}{3}+\cdots=C+\sum_{n=0}^{\infty}\left(\dfrac{c_n\left(x-x_0\right)^{n+1}}{n+1}\right)$

The radii of convergence of the power series in one and two are both *R*.

Note: This theorem indicates that we should be able to differentiate a power series infinitely many times!

<u>Example 4:</u> Use $f\left(x\right)=\dfrac{1}{1-x}=\sum_{n=0}^{\infty}x^n$ on $\left(-1,1\right)$ to express $\dfrac{1}{\left(1-x\right)^2}$ as a power series. State the radius of convergence of the series that you find.

We note that by the above theorem: $f'(x)=\dfrac{1}{(1-x)^2}=\sum_{n=1}^{\infty} nx^{n-1}$. If we want the index to start at zero (we don't *need* to have this) we can write: $\dfrac{1}{(1-x)^2}=\sum_{n=0}^{\infty}(n+1)x^n$. We know by the above theorem that the radius of convergence of this series is 1 since the radius of convergence of $\sum_{n=0}^{\infty} x^n$ is 1.

Example 5: Find a power series representation for $g(x)=\ln(1+x)$ and state its radius of convergence.

We first note that $g'(x)=\dfrac{1}{1+x}$. We calculate that $\dfrac{1}{1+x}=\dfrac{1}{(1-(-x))}$. We know that this is the sum of the geometric series with $a=1$ and $r=-x$ provided that $|-x|<1 \Leftrightarrow -1<x<1$.

This means that a power series representation for $\dfrac{1}{1+x}$ is: $\sum_{n=0}^{\infty}\left((-x)^n\right)=\sum_{n=0}^{\infty}\left((-1)^n x^n\right)$ and its radius of convergence is 1. So, we have $\dfrac{1}{1+x}=\sum_{n=0}^{\infty}\left((-1)^n x^n\right)$. By the most recent theorem we know that we can integrate both sides of this equation. When we do this we obtain:

$\ln(1+x)=C+\sum_{n=0}^{\infty}\left(\dfrac{(-1)^n x^{n+1}}{n+1}\right)$ (*Note: We only know for sure that this equation holds for values of x in* $(-1,1)$ *though it can be proven to hold when x equals 1 as well*). Now, to find C we choose a value of x for which we know $\ln(1+x)$ and which makes the series simple. The best choice is $x = 0$. When we plug this into the equation we obtain: $\ln(1)=C \Leftrightarrow C=0$. Thus, we have that $g(x)=\ln(1+x)=\sum_{n=0}^{\infty}\left(\dfrac{(-1)^n x^{n+1}}{n+1}\right)$ for x in $(-1,1)$ and the power series representation we obtained for $g(x)$ has radius of convergence equal to one.

Note: As mentioned above the final equation also holds for x equal to one. In this case we have $\ln(2)=\sum_{n=0}^{\infty}\left(\dfrac{(-1)^n}{n+1}\right)=1-\dfrac{1}{2}+\dfrac{1}{3}-\dfrac{1}{4}+\cdots$ *which is a result that was mentioned in a previous lecture.*

Example 6 (you try): Find a power series representation for $f(x) = \tan^{-1}(x)$ and state its radius of convergence.

We first note that $f'(x) = \frac{1}{1+x^2}$. We calculate that $\frac{1}{1+x^2} = \frac{1}{\left(1-\left(-x^2\right)\right)}$. We know that this is the sum of the geometric series with $a=1$ and $r=-x^2$ provided that $\left|-x^2\right| < 1 \Leftrightarrow -1 < x < 1$.

This means that a power series representation for $\frac{1}{1+x^2}$ is: $\sum_{n=0}^{\infty}\left(\left(-x^2\right)^n\right) = \sum_{n=0}^{\infty}\left((-1)^n x^{2n}\right)$ and its radius of convergence is 1. So, we have $\frac{1}{1+x^2} = \sum_{n=0}^{\infty}\left((-1)^n x^{2n}\right)$. By the most recent theorem we know that we can integrate both sides of this equation. When we do this we obtain:

$\tan^{-1}(x) = C + \sum_{n=0}^{\infty}\left(\frac{(-1)^n x^{2n+1}}{2n+1}\right)$ (*Note: We only know for sure that this equation holds for values of x in* $(-1,1)$ *though it can be proven to hold when x equals 1 and -1 as well*). Now, to find C we choose a value of x for which we know $\tan^{-1}(x)$ and which makes the series simple. The best choice is $x = 0$. When we plug this into the equation we obtain: $\tan^{-1}(0) = C \Leftrightarrow C = 0$. Thus, we have that $f(x) = \tan^{-1}(x) = \sum_{n=0}^{\infty}\left(\frac{(-1)^n x^{2n+1}}{2n+1}\right)$ for x in $(-1,1)$ and the power series representation we obtained for $g(x)$ has radius of convergence equal to one.

Note: As mentioned above the final equation also holds for x equal to one, in this case we have

$$\frac{\pi}{4} = \tan^{-1}(1) = \sum_{n=0}^{\infty}\left(\frac{(-1)^n}{2n+1}\right) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$